CREDIT RISK ANALYSIS SYSTEM

PROJECT REPORT

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Abstract
The Credit Risk Analysis System analyzes loan applicant information and estimates the possibility of loan default. The project uses a dataset of 5,000 applicant records. It includes data preparation, exploratory data analysis, SQL summaries, machine-learning model comparison, and an interactive Gradio dashboard. Logistic Regression, Decision Tree, and Random Forest are compared using classification metrics. In the recorded experiment, Decision Tree achieved the highest F1-score among the three tested models. The dashboard allows users to enter applicant details and view a predicted risk category and estimated default probability. This output is for educational decision support and should not be the sole basis for a real loan decision.

1. Introduction
Credit risk is the possibility that a borrower may not repay a loan as agreed. Reviewing applicant information can help identify patterns associated with default. This project demonstrates a beginner-friendly data analysis and machine-learning workflow for estimating loan default risk from historical applicant data.

2. Problem Statement
Reviewing many loan applications manually can be time-consuming. This project explores whether historical applicant attributes can be used to classify applications into predicted default and non-default categories and present summaries through a simple dashboard.

3. Objectives
- Load and prepare the credit-risk dataset.
- Explore applicant data and display patterns using graphs.
- Use SQL queries to summarize defaults and applicant categories.
- Train and compare Logistic Regression, Decision Tree, and Random Forest.
- Evaluate models using accuracy, precision, recall, F1-score, and ROC-AUC.
- Build a dashboard for dataset exploration, model comparison, and new-applicant prediction.

4. Dataset Description
The project uses a 5,000-row dataset with 12 columns. The fields describe applicant age, income, home ownership, employment length, loan purpose, loan grade, loan amount, interest rate, loan status, loan-to-income proportion, previous default history, and credit-history length.

Column	Description
person_age	Applicant age
person_income	Annual income
person_home_ownership	Home ownership category
person_emp_length	Employment length
loan_intent	Purpose of the loan
loan_grade	Loan grade in the source data
loan_amnt	Loan amount
loan_int_rate	Loan interest rate

loan_status	Target: 0 = non-default, 1 = default
loan_percent_income	Loan amount as a proportion of income
cb_person_default_on_file	Previous default indicator
cb_person_cred_hist_length	Credit-history length

5. Data Preparation

Data preparation was performed before modeling. Duplicate records and missing values were reviewed, and a cleaned dataset was saved. The saved model uses a preprocessing pipeline: numeric columns are imputed using the median and scaled; categorical columns are imputed using the most frequent value and one-hot encoded. The model input excludes loan_grade and uses the applicant attributes required by the trained pipeline.

Before submission, add the exact final missing-value handling and the number of records removed from the final notebook output, if detailed cleaning statistics are required.

6. Exploratory Data Analysis

The dashboard displays charts for loan-status distribution, applicant income distribution, loan-amount distribution, and home-ownership categories. These charts summarize the dataset and show how applicant records are distributed.

Dataset size: 5,000 applicants. Non-default cases: 3,611. Default cases: 1,389. Overall default rate: 27.78%.

7. SQL Analysis

SQLite was used to store and query the cleaned applicant data. SQL summaries include total applicants, default and non-default counts, default percentages by loan purpose, and home-ownership summaries. These reports complement the dashboard charts.

The recorded summary contains 5,000 applicants, 1,389 default cases, and 3,611 non-default cases, giving a default rate of 27.78%.

8. Machine-Learning Models

Three classification algorithms were trained and compared:

Logistic Regression: a linear classification model used as a baseline.

Decision Tree: a tree-based model that learns decision rules from applicant attributes.

Random Forest: an ensemble model that combines multiple decision trees.

The saved model is a preprocessing pipeline with a DecisionTreeClassifier using class_weight='balanced', max_depth=10, and random_state=42.

9. Model Evaluation and Comparison

The table below shows the recorded results. Decision Tree achieved the highest F1-score among the three models. Accuracy alone is not enough to evaluate default detection, so precision, recall, F1-score, and ROC-AUC are also reported.

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Logistic Regression	0.726	0.5263	0.1439	0.2260	0.6398
Decision Tree	0.604	0.3370	0.4388	0.3813	0.5417
Random Forest	0.704	0.4118	0.1511	0.2211	0.5964

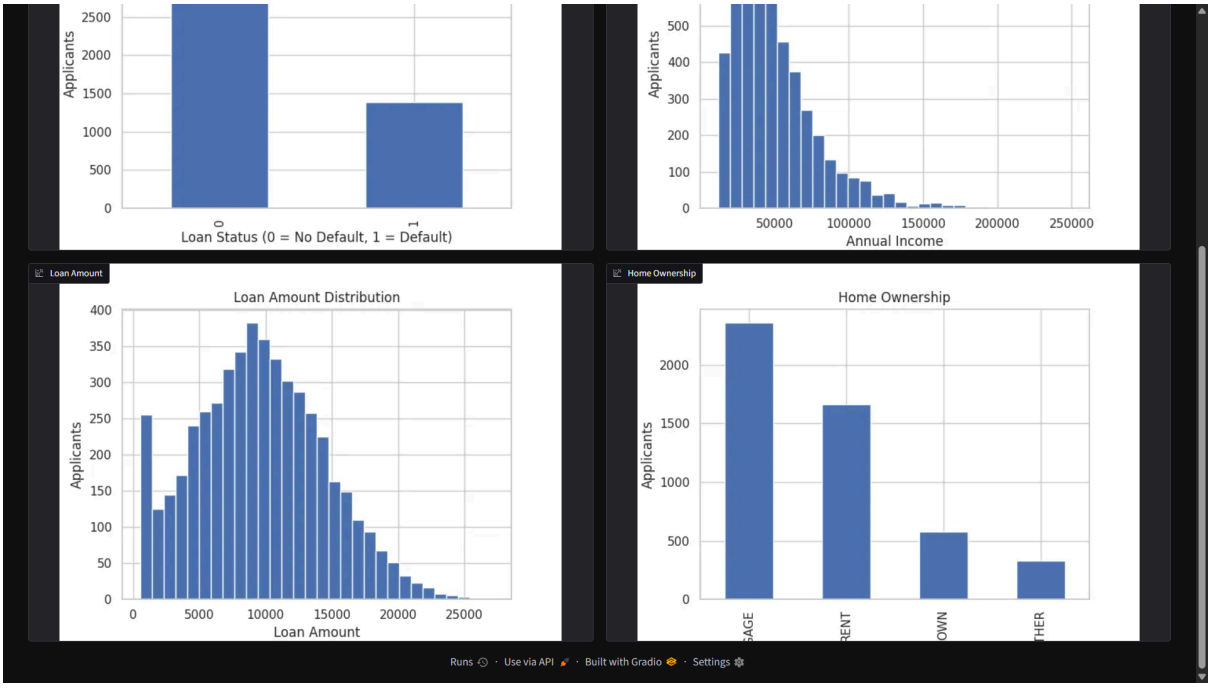
Decision Tree was selected because it achieved the highest F1-score (0.3813) and recall (0.4388) in the recorded comparison. The scores also indicate that the model has limitations and needs further validation before any real-world use.

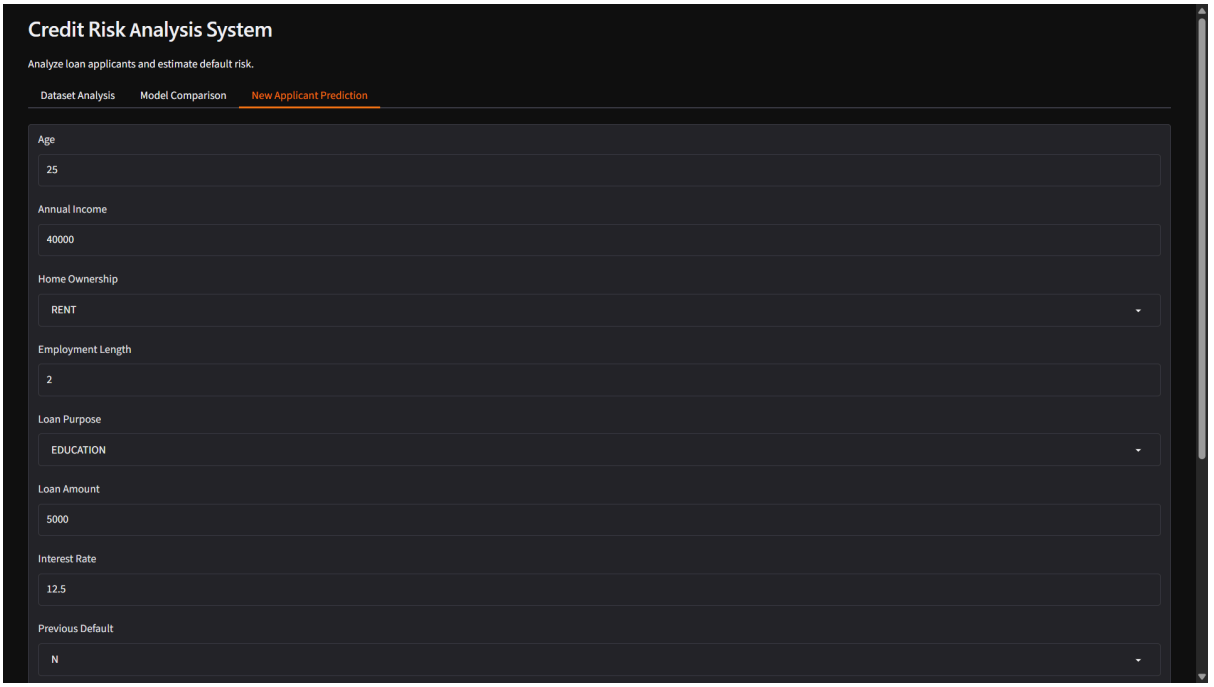
10. Applicant Prediction Dashboard

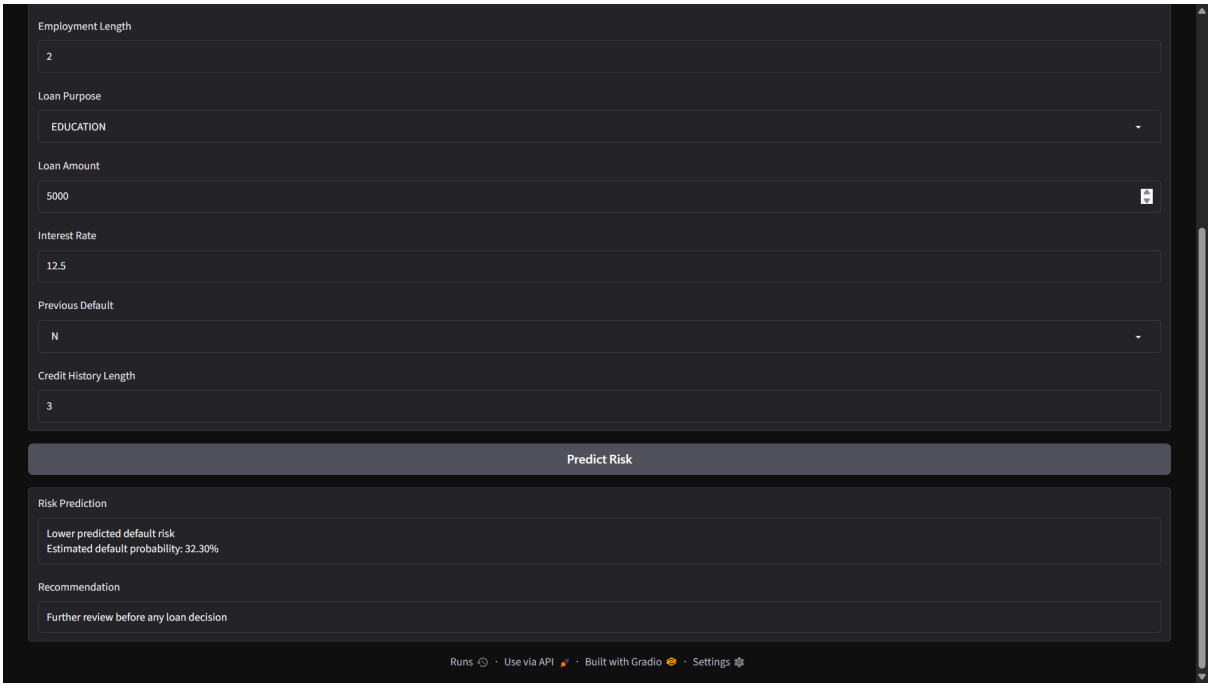
The Gradio dashboard contains three sections: Dataset Analysis, Model Comparison, and New Applicant Prediction. Users can enter age, annual income, home ownership, employment length, loan purpose, loan amount, interest rate, previous default indicator, and credit-history length. The system returns a predicted risk category and estimated default probability. The interface is intended for demonstration and education.

11. Results and Discussion

- The dataset contains 5,000 applicant records.
- There are 1,389 default cases (27.78%) and 3,611 non-default cases.
- Decision Tree achieved the highest F1-score among the tested models.
- SQL reports and dashboard charts help users inspect dataset patterns.
- The prediction interface demonstrates how a trained model can be applied to a new applicant record.







12. Limitations

- Model results depend on the quality and representativeness of the dataset.
- The F1-score and ROC-AUC indicate that prediction performance is limited.
- A prediction is not a guarantee that an applicant will or will not default.
- The Gradio public share link may be temporary and may stop working when the runtime ends.
- The system should not be the sole basis for a real lending decision.

13. Future Enhancements

- Test additional algorithms and tune model parameters.
- Improve recall and F1-score while reviewing false positives and false negatives.

- Add more visualizations and a downloadable prediction report.
- Deploy the dashboard on a stable hosting platform.
- Test on more recent data and check for unfair outcomes across applicant groups.

14. Conclusion

The Credit Risk Analysis System combines data preparation, exploratory analysis, SQL reporting, model comparison, and an interactive Gradio dashboard. Among the three tested classifiers, Decision Tree achieved the highest F1-score in the recorded experiment and was saved for applicant prediction. The project demonstrates an end-to-end workflow and highlights the importance of careful evaluation, validation, and responsible use of predictive results.